

John Marsden

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LinkedIn

EDUCATION	North Carolina State University	Graduation Date: 4/30/25
	Ph.D. – Computer Science, GPA: 3.81	
	Abilene Christian University	Graduation Date: 4/30/20
	B.S. – Math and Computer Science, GPA: 3.84	
SKILLS	Languages: C#, Python, Java, SQL, Rust, HTML, CSS, JavaScript, F#, Lua, C++, C, Ruby, Haskell Frameworks: Pytorch, Tensorflow, ASP.NET, Rails, Vue Databases: Oracle Database, Postgres, MySQL Dev Ops: Fuzzing, Github Actions, Docker, Bash, Linux Machine Learning: NLP, Attention Networks, Neural Networks, Graph Neural Networks	
EXPERIENCE	Instructor - Software Development Fundamentals NCSU	5/31/22 – 7/31/22
	◦ Taught the software development process and test-driven development to 20 students ◦ Engaged students through question and answer sections in each lecture ◦ Created and administered three exams to validate students' learning progress	
	Instructor - Introduction to Computing NCSU	12/31/21 – 4/30/22
	◦ Taught the fundamentals of Java to 40 students ◦ Built two autograders using JUnit 5 to autograde student code submissions ◦ Designed and administered three exams to validate students' learning progress	
	Teaching Assistant NCSU	7/31/20 – Present
	◦ Helped 300+ students navigate the complexities of multi-project Java codebases ◦ Validated 600+ student algorithmic analyses and algorithm designs ◦ Graded 300+ student algorithms using common data structures, including stacks, graphs, etc.	
PROJECTS	Machine Learning on Intermediate Languages	Invalid Date – Present
	◦ Created a code embedding machine learning model to convert control-flow graphs (CFG) to vectors ◦ Wrote a custom intermediate language representation to represent CFGs for embedding ◦ Reimplemented the deepwalk learning algorithm for distributed learning ◦ Constructed an attention network for using CFG embeddings to predict student performance	
	General Purpose Discord Bot	Invalid Date – Present
	◦ Built Norm: A general-purpose Discord bot written with the .NET runtime ◦ Implemented an HTML to Discord Markdown transpiler ◦ Built an event announcement system that supported timezone adjustment and role-based announcements ◦ Created an OMDb integration to allow for users to plan movie nights	